LAB 7

Perform Nested and Complex Queries

Aim: Mastering Subqueries (IN, NOT IN, EXISTS, NOT EXISTS, ANY, ALL), Correlated Subqueries, Derived Tables (FROM clause), HAVING clauses, and Set Operators (UNION, INTERSECT).

1 [Subquery - IN]: Write an SQL query using an IN subquery to find the full names and countries of players who scored more than 50 runs (NRuns > 50) in any single match.

```
SELECT CONCAT(FName, ' ', LName) AS FullName, Country
FROM player
WHERE PlayerID IN (
   SELECT PID
   FROM batting
   WHERE NRuns > 50
);
```

Schema: d6_10

FullName	Country
MS Dhoni	India
Sanath Jayasuriya	Sri Lanka
Sachin Tendulkar	India

2 [Subquery - NOT IN]: Retrieve all players (PlayerID, FName, LName, Country) who have NEVER bowled in any match using a NOT IN subquery referencing the Bowling table.

```
SELECT PlayerID, FName, LName, Country

FROM player

WHERE PlayerID NOT IN (

   SELECT PID

   FROM bowling

);
```

Schema: d6_10

PlayerID	FName	LName	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia

Schema: d6_10

PlayerID	FName	LName	Country
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan
99002	Adam	Gilchrist	Australia
99003	Michale	Vaughan	England
300311	Michale	Clark	NULL
406317	Sam	NULL	India
NULL	NULL	NULL	NULL

3 [Subquery - EXISTS]: Write an SQL query using EXISTS to list all matches (MatchID, Ground, Date) where at least one player hit 2 or more sixes (Six >= 2).

```
SELECT M.MatchID, M.Ground, M.Date
FROM odimatch AS M
WHERE EXISTS (
   SELECT 1
   FROM batting AS B
   WHERE B.MatchID = M.MatchID
    AND B.Six >= 2
```

);

Schema: d6_10

MatchID	Ground	Date
2755	Colombo	2008-08-24
2689	Brisbane	2008-03-04
NULL	NULL	NULL

4 [Subquery - NOT EXISTS]: Retrieve all players (PlayerID, FName, LName) for whom NO bowling record exists in the Bowling table using a correlated NOT EXISTS clause.

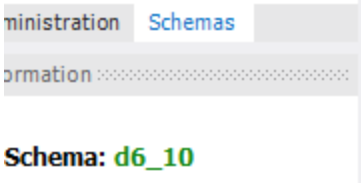
```
SELECT P.PlayerID, P.FName, P.LName
FROM player AS P
WHERE NOT EXISTS (
   SELECT 1
   FROM bowling AS B
   WHERE B.PID = P.PlayerID
);
```

ministration Schemas

ormation

Schema: d6_10

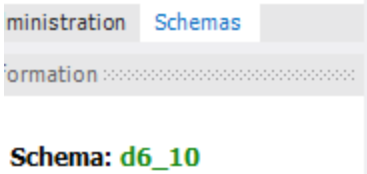
PlayerID	FName	LName
21001	Virender	Sehwag
25001	MS	Dhoni
27001	Praveen	Kumar
27002	Ishant	Sharma
89001	Sachin	Tendulkar
90001	Brain	Lara
90002	Anil	Kumble
92001	Shane	Warne
92003	Inzamam	UI-Huq
93002	Heath	Streak
93003	Gary	Kirsten
94004	Stephen	Fleming
95001	Ricky	Ponting
95002	Shaun	Pollock
95003	Jacques	Kallis
96001	Rahul	Dravid
96002	Herschelle	Gibbs
96003	Saurav	Ganguly
97004	Daniel	Vettori
98001	Shahid	Afridi
99002	Adam	Gilchrist



PlayerID	FName	LName
89001	Sachin	Tendulkar
90001	Brain	Lara
90002	Anil	Kumble
92001	Shane	Warne
92003	Inzamam	UI-Huq
93002	Heath	Streak
93003	Gary	Kirsten
94004	Stephen	Fleming
95001	Ricky	Ponting
95002	Shaun	Pollock
95003	Jacques	Kallis
96001	Rahul	Dravid
96002	Herschelle	Gibbs
96003	Saurav	Ganguly
97004	Daniel	Vettori
98001	Shahid	Afridi
99002	Adam	Gilchrist
99003	Michale	Vaughan
300311	Michale	Clark
406317	Sam	NULL
NULL	NULL	NULL

5 [Scalar Subquery in WHERE]: Find all players (FName, LName, Yborn) who were born before the average birth year (Yborn < AVG(Yborn)) of all registered players.

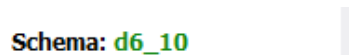
```
SELECT FName, LName, Yborn
FROM player
WHERE Yborn < (
   SELECT AVG(Yborn)
   FROM player
);
```



FName	LName	Yborn
Sachin	Tendulkar	1973
Brain	Lara	1969
Anil	Kumble	1970
Sanath	Jayasuriya	1969
Shane	Warne	1969
Muthiah	Murlitharan	1972
Inzamam	UI-Huq	1970
Heath	Streak	1974
Gary	Kirsten	1967
Chaminda	Vaas	1974
Stephen	Fleming	1973
Ricky	Ponting	1974
Shaun	Pollock	1973
Rahul	Dravid	1973
Herschelle	Gibbs	1974
Saurav	Ganguly	1972
Adam	Gilchrist	1971
Michale	Vaughan	1974

6 [Subquery - ALL Operator]: Write a query to find player batting performances where NRuns is greater than ALL runs scored by players representing 'Australia'.

```
SELECT B.*
FROM batting AS B
WHERE B.NRuns > ALL (
   SELECT BT.NRuns
   FROM batting AS BT
   JOIN player AS P
    ON BT.PID = P.PlayerID
   WHERE P.Country = 'Australia'
);
```



MatchID	PID	POrder	HOut	FOW	NRuns	Mts	Nballs	Fours	Six
2755	25001	5	C	232	71	104	80	4	0
2755	91001	1	C	74	60	85	52	8	2
2689	89001	2	C	205	91	176	121	7	0

7 [Subquery - ANY / SOME]: Find all bowling performances where Nwickets is greater than ANY wicket count achieved by bowlers from 'Sri Lanka'.

```
SELECT B.*
FROM bowling AS B
WHERE B.Nwickets > ANY (
   SELECT BW.Nwickets
   FROM bowling AS BW
   JOIN player AS P
     ON BW.PID = P.PlayerID
   WHERE P.Country = 'Sri Lanka'
);
```

Schema: d6_10

MatchID	PID	NOvers	Maidens	Nruns	Nwickets
2689	99001	10	0	58	1
2689	24001	3	0	27	1
2755	94002	9	1	40	1
2755	92002	10	0	56	1
2755	23001	10	0	53	2
2755	98002	10	0	40	3
2689	98002	10	0	44	1

8 [Correlated Subquery]: Write a correlated subquery to list each player who scored the maximum runs (NRuns) among all players from their own Country.

```
SELECT DISTINCT
   P.PlayerID,
   P.FName,
   P.LName,
   P.Country,
   B.NRuns
FROM player AS P
```

```
JOIN batting AS B
  ON P.PlayerID = B.PID
WHERE B.NRuns = (
    SELECT MAX(BT.NRuns)
    FROM player AS P2
    JOIN batting AS BT
      ON P2.PlayerID = BT.PID
    WHERE P2.Country = P.Country
);
```

Schema: d6_10

PlayerID	FName	LName	Country	NRuns
91001	Sanath	Jayasuriya	Sri Lanka	60
89001	Sachin	Tendulkar	India	91
24001	Andrew	Symonds	Australia	42

9 [Nested Subquery with HAVING]: Display players whose average runs per match is strictly greater than the overall average runs scored across all matches in the Batting table.

```
SELECT
    P.PlayerID,
    P.FName,
    P.LName,
    AVG(B.NRuns) AS AverageRuns
FROM player AS P
JOIN batting AS B
  ON P.PlayerID = B.PID
GROUP BY P.PlayerID, P.FName, P.LName
HAVING AVG(B.NRuns) > (
```

```
    SELECT AVG(NRuns)
    FROM batting
);
```

Schema: d6_10

PlayerID	FName	LName	AverageRuns
25001	MS	Dhoni	53.5000
91001	Sanath	Jayasuriya	60.0000
89001	Sachin	Tendulkar	91.0000
24001	Andrew	Symonds	42.0000

10 [Derived Table in FROM]: Create a derived table in the FROM clause calculating total runs and total balls faced per player, then select players with a strike rate ((Total_Runs / Total_Balls) * 100) greater than 100.

```
SELECT
    P.PlayerID,
    P.FName,
    P.LName,
    D.Total_Runs,
    D.Total_Balls,
    (D.Total_Runs / D.Total_Balls) * 100 AS Strike_Rate
FROM player AS P
JOIN (
    SELECT
        PID,
        SUM(NRuns) AS Total_Runs,
        SUM(Nballs) AS Total_Balls
    FROM batting
    GROUP BY PID
) AS D
```

```
 ON P.PlayerID = D.PID
WHERE D.Total_Balls > 0
 AND (D.Total_Runs / D.Total_Balls) * 100 > 100;
```

Schema: d6_10

PlayerID	FName	LName	Total_Runs	Total_Balls	Strike_Rate
91001	Sanath	Jayasuriya	60	52	115.3846

11 [Set Operator - UNION]: Write a query using UNION to combine the PlayerIDs of players who scored > 50 runs in Batting with players who took >= 3 wickets in Bowling.

```
SELECT PID AS PlayerID
FROM batting
WHERE NRuns > 50
UNION
SELECT PID AS PlayerID
FROM bowling
WHERE Nwickets >= 3;
```

Schema: d6_10

PlayerID
25001
91001
89001
98002

12 [Set Operator / Subquery Intersect]: Find players who both batted AND bowled in MatchID = 2755 using an INTERSECT query or equivalent subquery structure.

```
SELECT PlayerID, FName, LName
FROM player
WHERE PlayerID IN (
   SELECT PID
   FROM batting
```

```
    WHERE MatchID = 2755
)
AND PlayerID IN (
    SELECT PID
    FROM bowling
    WHERE MatchID = 2755
);
```

Schema: d6_10

PlayerID	FName	LName
23001	Yuvraj	Singh
91001	Sanath	Jayasuriya
94002	Chaminda	Vaas
92002	Muthiah	Murlitharan
98002	Harbhajan	Singh

13 [Complex Grouping & HAVING]: Display Ground, total runs scored on that ground, and total matches played, considering only grounds where more than 2 matches took place and total runs exceed 100.

```
SELECT
    M.Ground,
    SUM(B.NRuns) AS Total_Runs,
    COUNT(DISTINCT M.MatchID) AS Total_Matches
FROM odimatch AS M
JOIN batting AS B
  ON M.MatchID = B.MatchID
GROUP BY M.Ground
HAVING COUNT(DISTINCT M.MatchID) > 2
   AND SUM(B.NRuns) > 100;
```

Schema: d6_10

Ground	Total_Runs	Total_Matches
Colombo	153	3